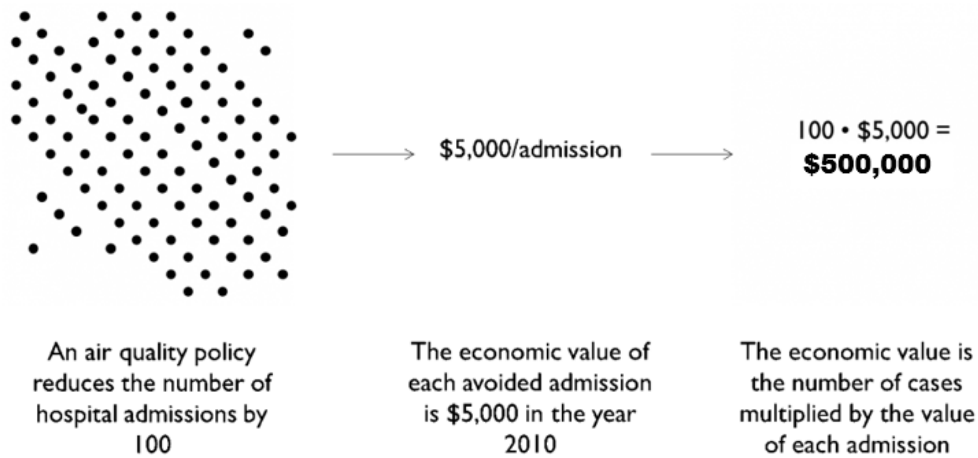

Figure 1-2. Estimating the Economic Value of Human Health Effects

There are several different ways of calculating the value of the health effect. For example, the value of an avoided premature mortality is generally calculated using the Value of Statistical Life (VSL). The value of a statistical life is the monetary value that a group of people are willing to pay to slightly reduce the risk of premature death in the population. For other health effects, the medical costs of the illness may be the only valuation data available. The BenMAP-CE database includes several different functions for VSL and valuation functions for other health effects for you to choose, or you can use the U.S. EPA's approach for quantifying and valuing air pollution effects[3].

Figure 1-3 summarizes the BenMAP-CE inputs and outputs. This figure shows the types of choices that you make regarding the modeling of population exposure, the types of health effects to model, and how to place an economic value on these health effects. Please note that BenMAP-CE does not have air quality modeling capabilities, and therefore the user must provide externally created data in order to work with modeled data. BenMAP-CE is preloaded with limited air quality monitoring data, but externally created monitoring data may also be needed.

What else can BenMAP-CE do?

BenMAP-CE incorporates a geographic information system (GIS), allowing users to create, utilize, visualize, and export maps of air pollution, population, incidence rates, incidence rate changes, economic valuations, and other types of data (Figure 1-4).

[3] See https://www.epa.gov/benmap/benmap-community-edition.

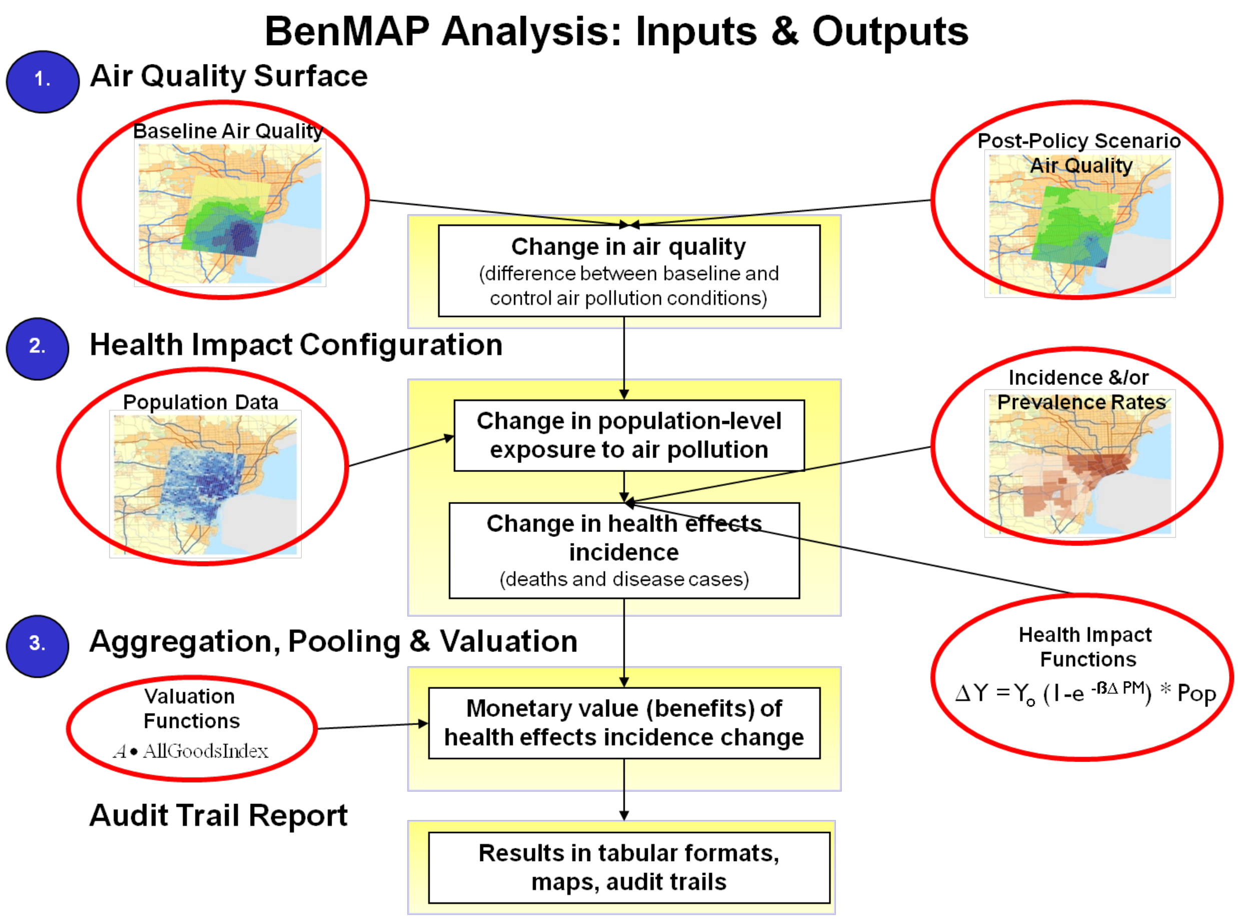

Figure 1-3. BenMAP-CE Flow Diagram

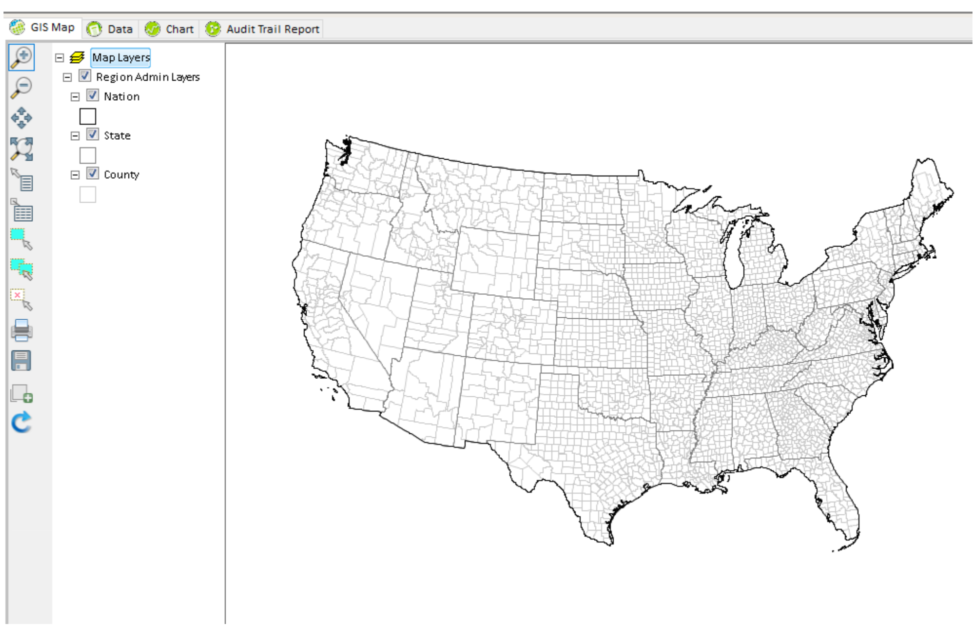

Figure 1-4. BenMAP-CE GIS Example

Analysts can use BenMAP-CE to:

- Create maps illustrating the population/community level ambient pollution levels;
- Compare benefits associated with various regulatory programs;
- Characterize the distribution of health impacts among population sub-groups;
- Estimate health impacts and economic values of existing air pollution concentrations;
- Estimate the health benefits of alternative ambient air quality standards; and
- Perform sensitivity analyses of health or valuation functions, or of other inputs.

Example Applications

Analysts have used BenMAP-CE to investigate a variety of policy questions such as:

- What is the current health burden from $PM_{2.5}$ levels in Addis Ababa?
- How large are the economic benefits of reduced maternal exposure to fine particulate matter?
- What are the future health impacts of wildfire smoke health impacts under meteorological scenarios?
- What are the benefit impacts of alternative air quality strategies in Detroit, MI?
- How large are tree and forest effects on air quality and human health?
- What are the health benefits from vehicular pollution control strategies?

1.2 How to Use this Manual

Chapters 2 through 9 of this manual provide step-by-step instructions on how to use BenMAP-CE. New users should start with Chapters 2 and 3, which are both relatively short. These chapters provide a basic overview of the tool and how it works, and explain some potentially confusing terminology.[4] Use the rest of the manual to answer any specific questions you may have, or to walk you step-by-step through the various components. Chapter 4 discusses how to enter data into BenMAP-CE, Chapters 5 through 7 cover each of the main steps in the Core Program, and Chapters 8 and 9 cover mapping, report options, and additional tools.

Each chapter is introduced by a short section that describes what you can find within the chapter and provides an outline of the chapter's contents. This is a good place to go if the Table of Contents does not provide enough detail for you to find the section you need. The end of most chapters has a series of "Frequently Asked Questions," which may also be helpful for answering specific questions. In chapters that provide instructions on navigating the tool, the following conventions are observed: tree menu items, buttons, tabs and selection box labels are in bold type; prompts and messages are enclosed in quotation marks; and drop-down menu items, options to click or check, and items that need to be filled in or selected by the user are italicized. Throughout the chapters you will also see boxes presenting common mistakes and important things to remember when working with BenMAP-CE.

[4] Another good reference is the BenMAP-CE Quick Start Guide, see: http://www.epa.gov/benmap.

There is also a set of Technical Appendices to provide more detailed information on model functions, data, and underlying assumptions.

Appendix A: Monitor Rollback Algorithms
Appendix B: Air Pollution Exposure Estimation Algorithms
Appendix C: Deriving Health Impact Functions
Appendix D: Health Incidence & Prevalence Data in U.S. Setup
Appendix E: Core Particulate Matter Health Impact Functions in U.S. Setup
Appendix F: Core Ozone Health Impact Functions in U.S. Setup
Appendix G: Additional Health Impact Functions in U.S. Setup
Appendix H: Core Health Valuation Functions in U.S. Setup
Appendix I: Additional Health Valuation Functions in U.S. Setup
Appendix J: Population & Other Data in U.S. Setup
Appendix K: Uncertainty & Pooling
Appendix L: Command Line BenMAP-CE
Appendix M: Function Editor
References

1.3 Computer Requirements

The computer hardware requirements for BenMAP-CE are typically modest, though this will vary depending on the complexity of the analysis. BenMAP-CE requires a Windows platform and can be used on machines running Windows 7, Windows 8 or Windows 10. In particular, BenMAP-CE requires a computer with:

- Either a 64- or 32-bit operating system, although a 64-bit operating system is recommended
- Adobe Acrobat Reader
- Microsoft Excel or other spreadsheet program (in order to read exported .xlsx files)[5]
- Microsoft .NET Framework 4 (or 4.5)[6]

[5] OpenOffice and LibreOffice are two open-source options for spreadsheet tools.

[6] If .NET is not pre-installed, BenMAP-CE will provide a message advising you to install .NET. A standalone installer is available on the Microsoft website (URL: http://www.microsoft.com/en-us/download/details.aspx?id=17718). Install the runtime version and associated files (e.g., dotnetfx40_full_x86_x64.exe).